Engine Model(s): PT6A

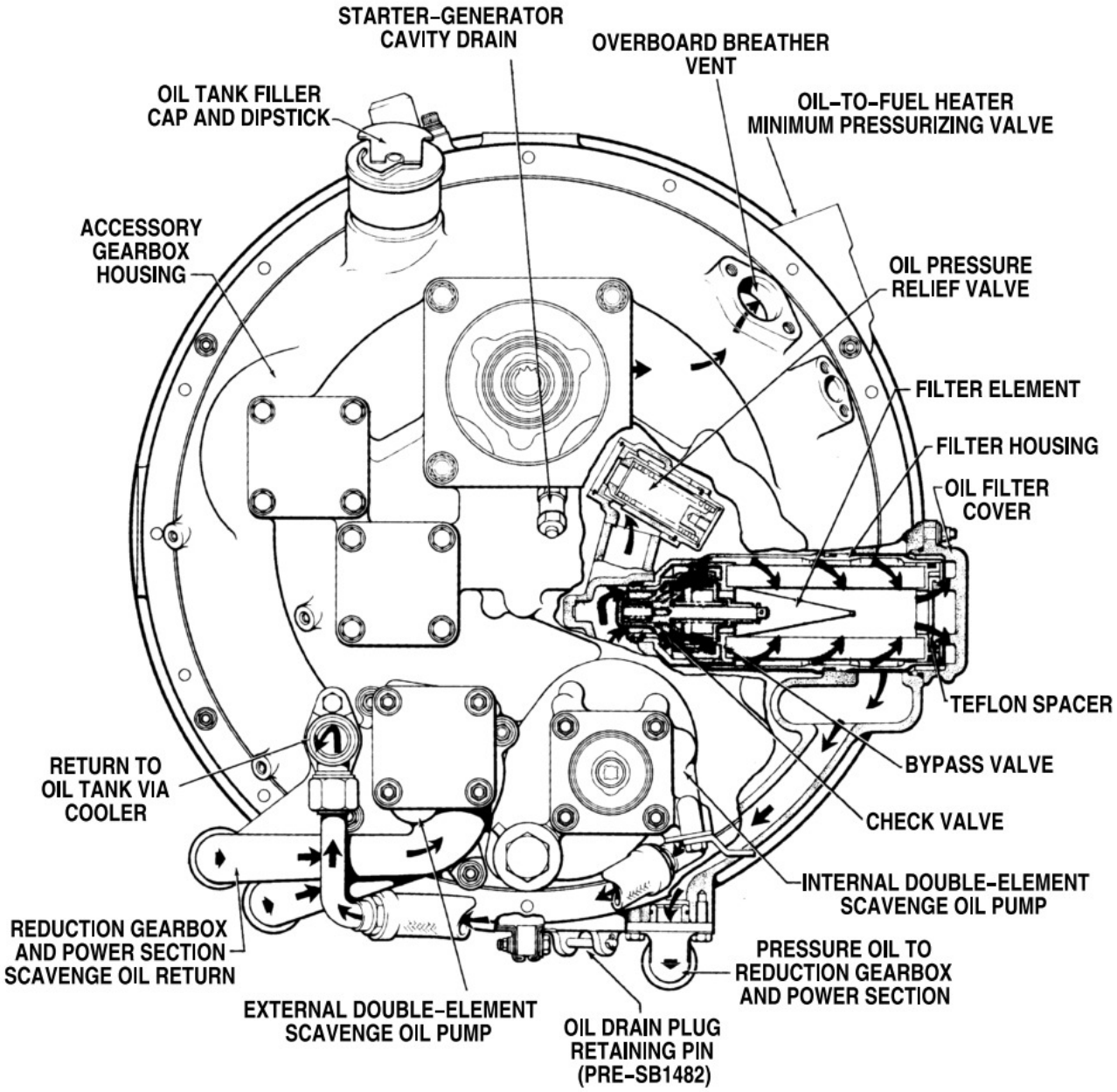
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OIL FILTER, HOUSING AND CHECK VALVE - DESCRIPTION AND OPERATION

1. Description and Operation (Ref. Fig. 1)

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Figure 1 Oil Filter and Pressure Relief Valve



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The oil filter assembly consists of a removable alloy filter housing containing either a disposable cartridge-type filter element or a permanent cleanable element, a spring-loaded bypass valve and a check valve. The oil filter assembly and filter housing is located in the compressor air inlet case at the 3 o'clock position and retained by the filter cover; the cover is fastened to the inlet case by self-locking nuts. The inner end of the filter housing locates in the oil pump support boss and is sealed by a preformed packing.

Pressure oil from the oil pump passes through holes in the housing, lifts the check valve off its seat and flows into the filter housing. Oil then filtrates through the filter element into the center core and out into cored passages in the inlet case. Any extraneous foreign matter in the pressure oil is deposited on the outer face of the filter element. The check valve closes under the influence of a spring to prevent gravity oil flow into the engine after shutdown and also facilitates filter element replacement without having to drain the oil tank.

A filter bypass is provided by a spring-loaded valve piston installed between the check valve and the inner end of the filter element. The bypass valve is normally closed and sealed from the element core. In the event of a filter blockage, increased pressure overcomes the bypass valve spring and moves the valve piston, enabling oil to flow directly into the center core of the filter element. On some filter elements, a coarse hat-type screen attached to the inner end of the filter element partially filters the bypass oil before entering the engine.

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